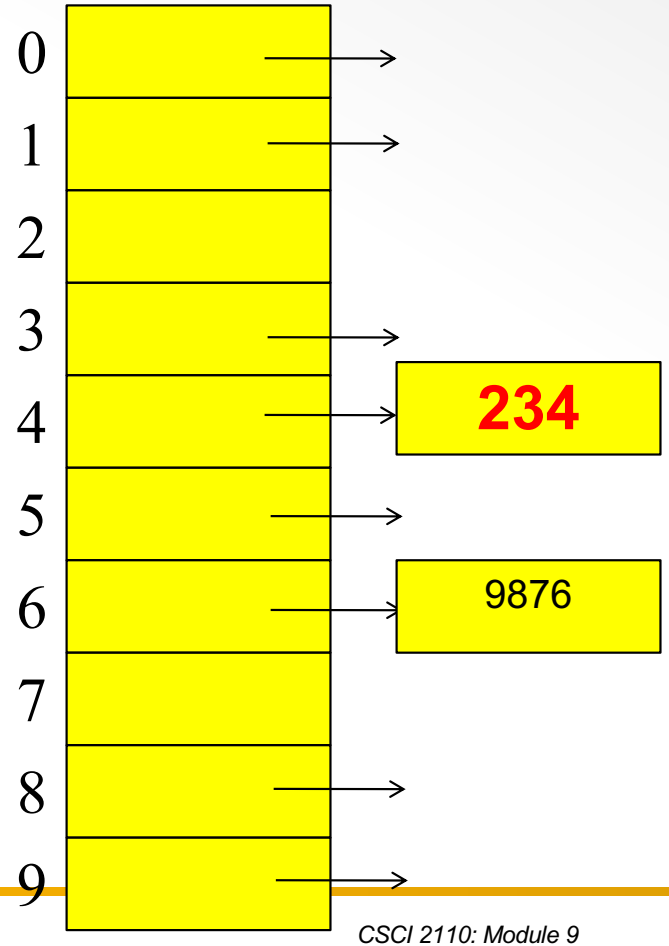
Open Hashing or Separate chaining

Concept: Create an ArrayList of Linked Lists



Keys:

9876, 234, 5672, 89990, 6777, 366, 7446, 212, 4676

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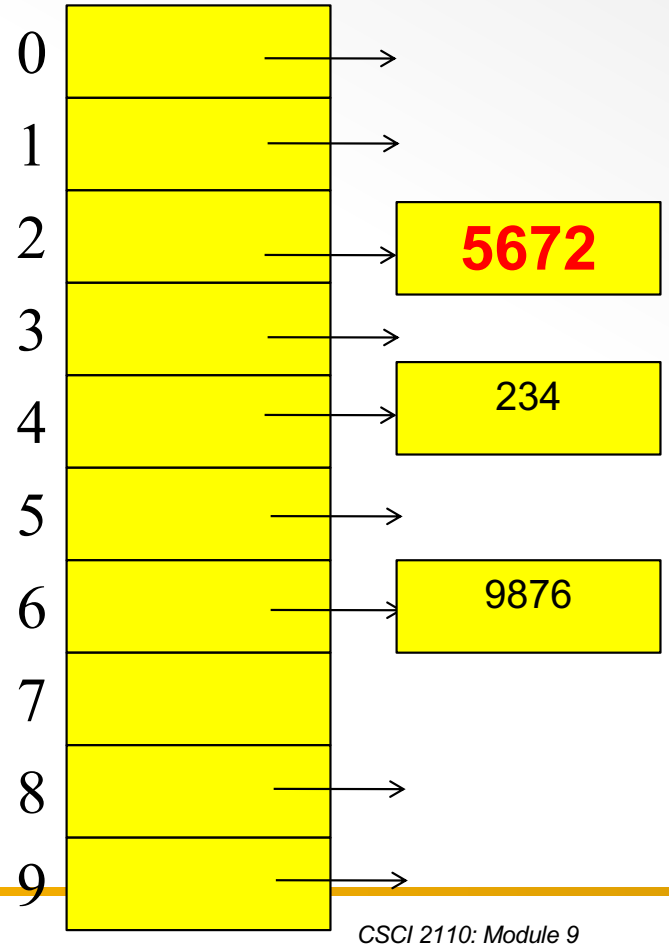

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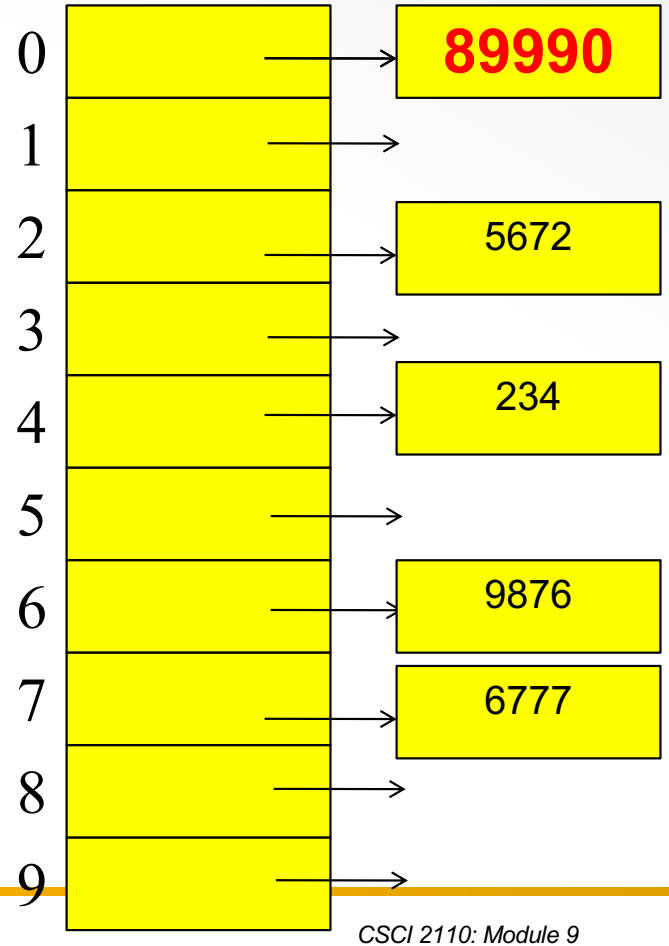

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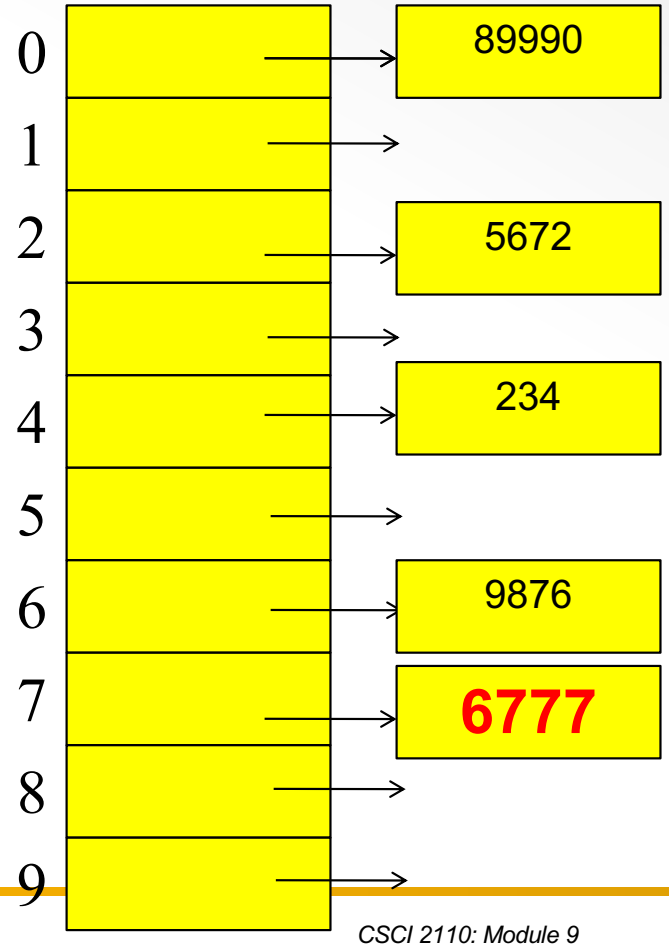

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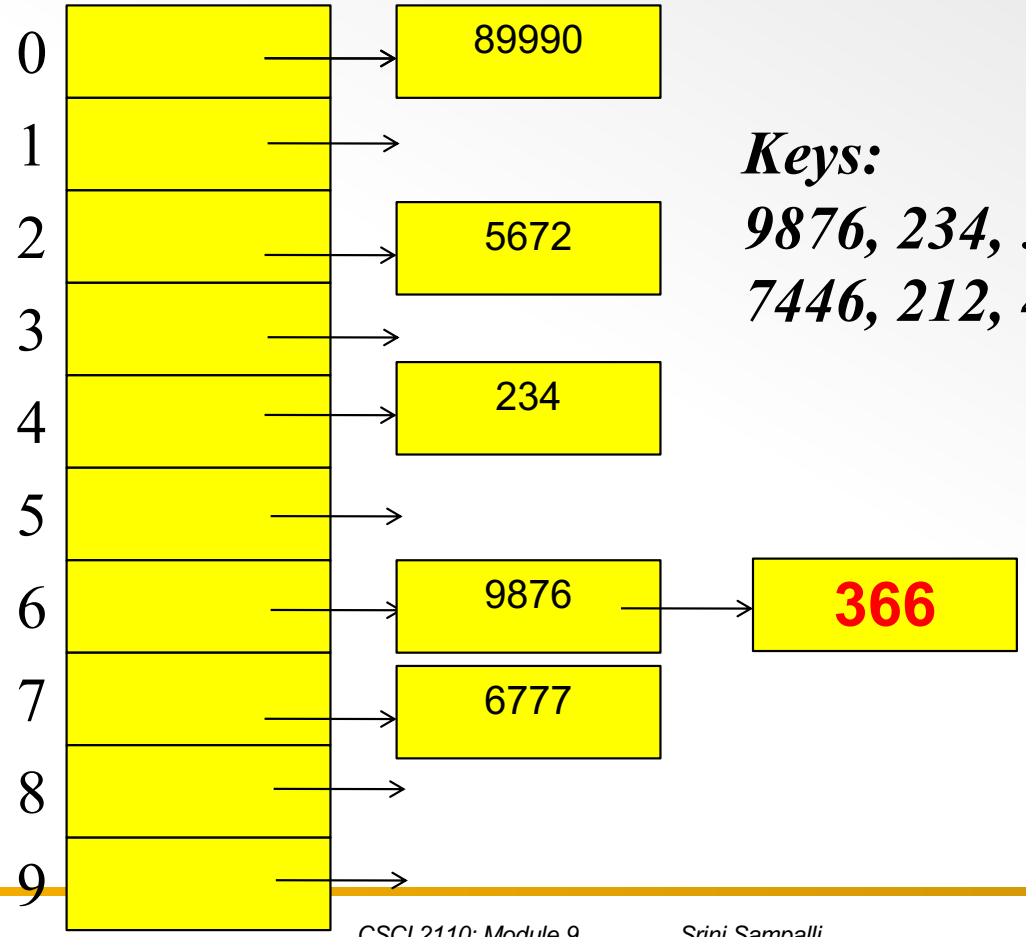

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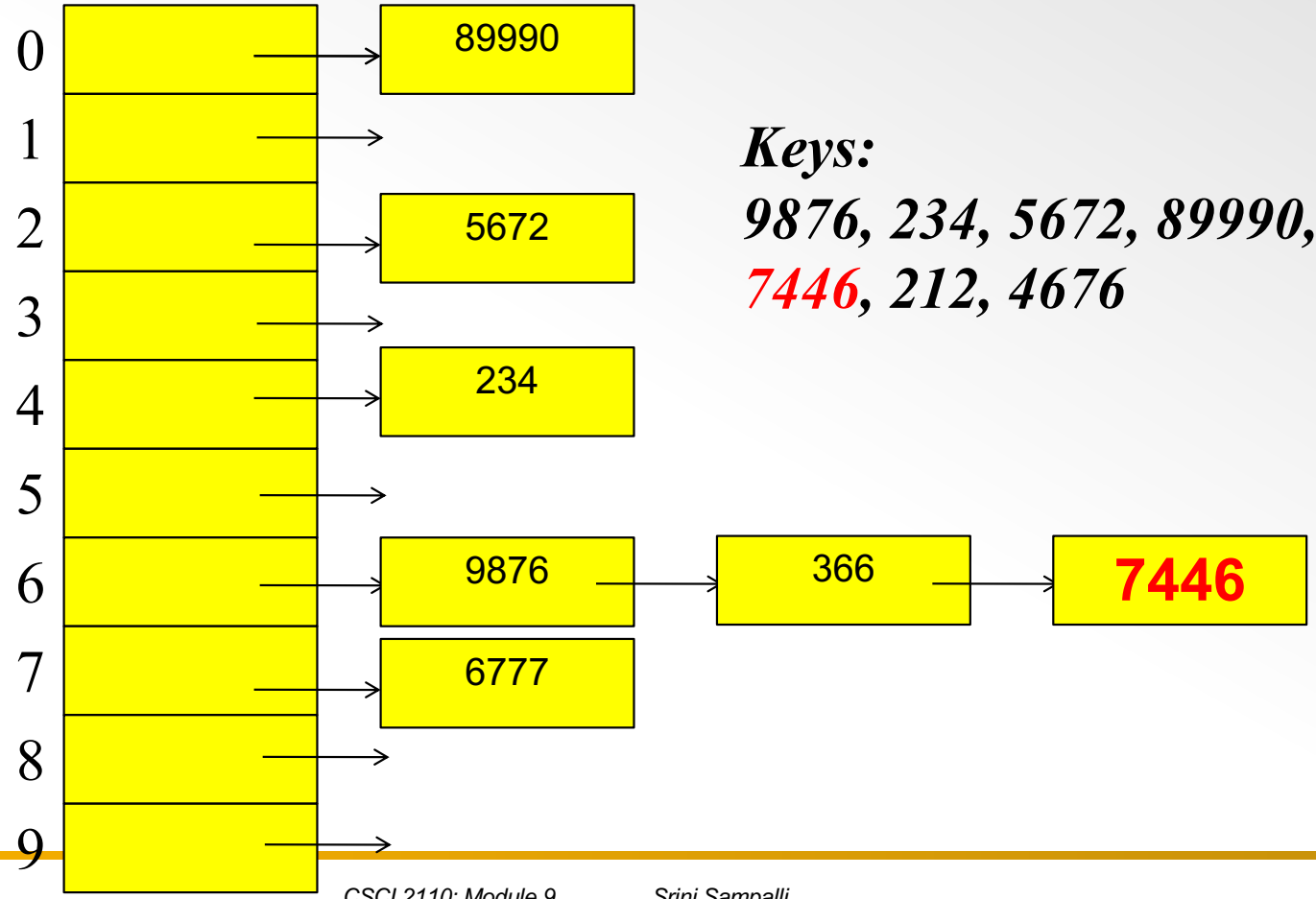

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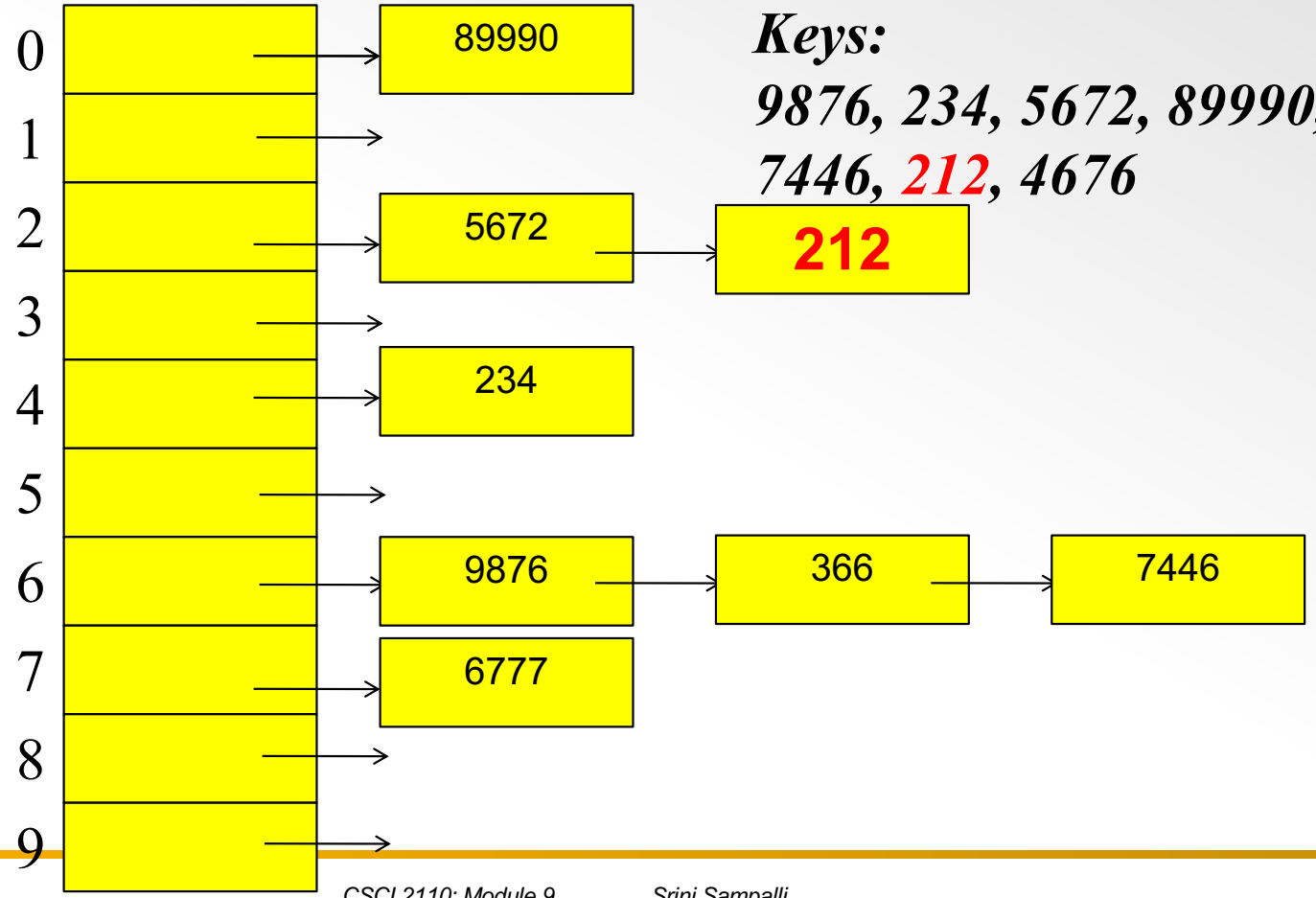

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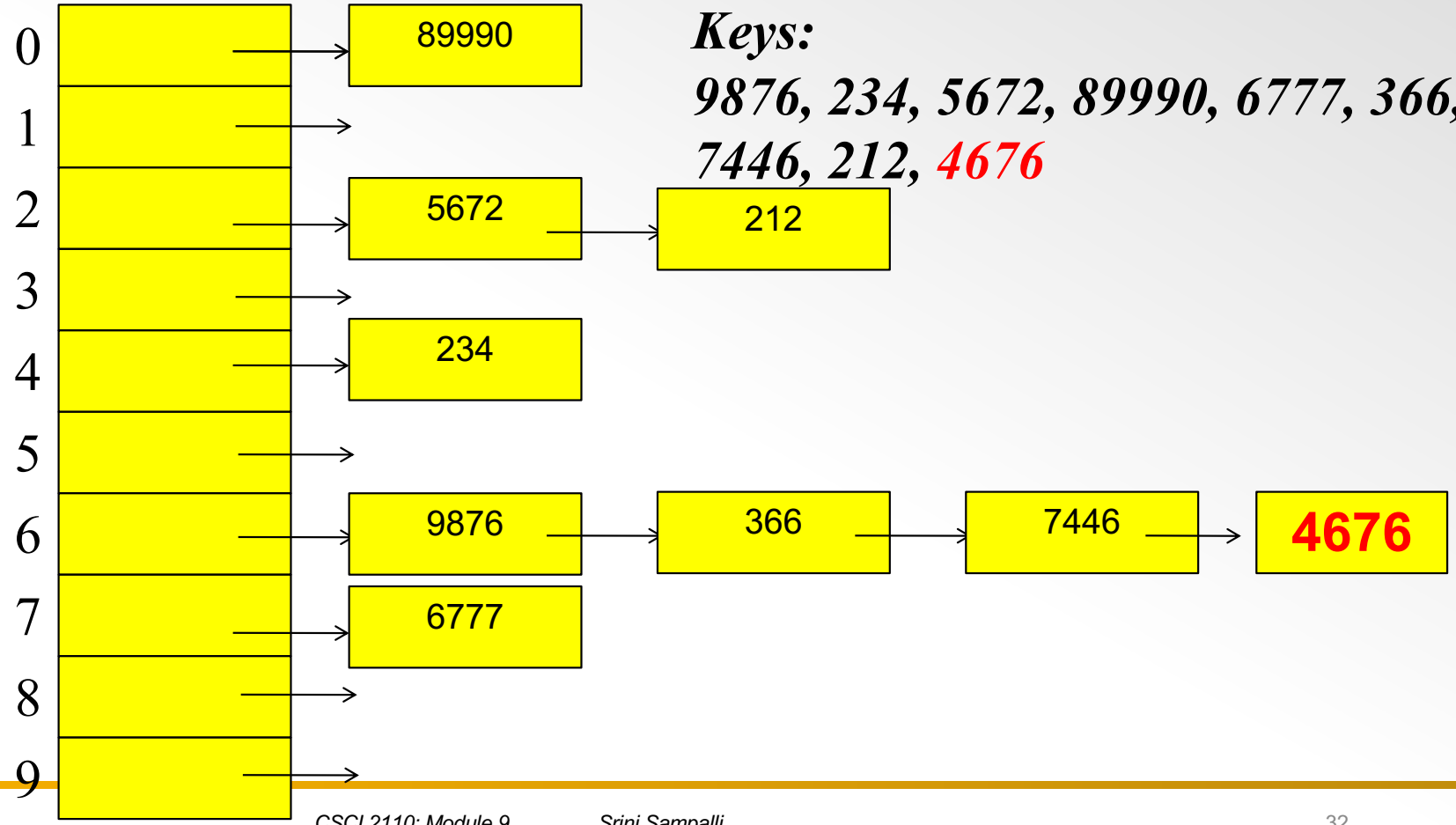

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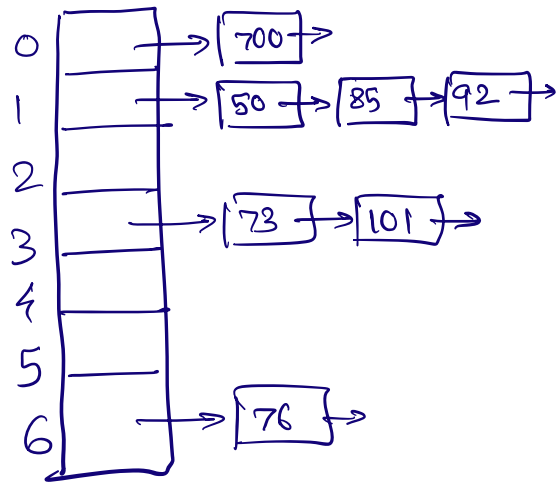
MORE EXAMPLES WITH OPEN HASHING

1. **Draw the hash table with open hashing resulting from the insertion of the following sequence of keys**
   50, 700, 76, 85, 92, 73, 101
   Assume that the size of the table is 7 and the hash function is key mod 7 mod %

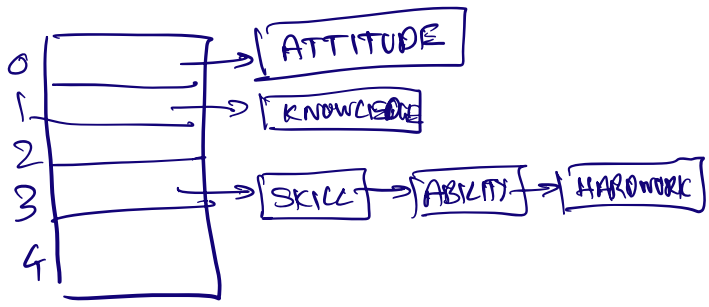


50 mod 7 = 1
700 mod 7 = 0
76 mod 7 = 6
85 mod 7 = 1
92 mod 7 = 1
73 mod 7 = 3
101 mod 7 = 3

2. Draw the hash table with open hashing resulting from the insertion of the following sequence of keys (words):
SKILL, ABILITY, KNOWLEDGE, HARDWORK, ATTITUDE

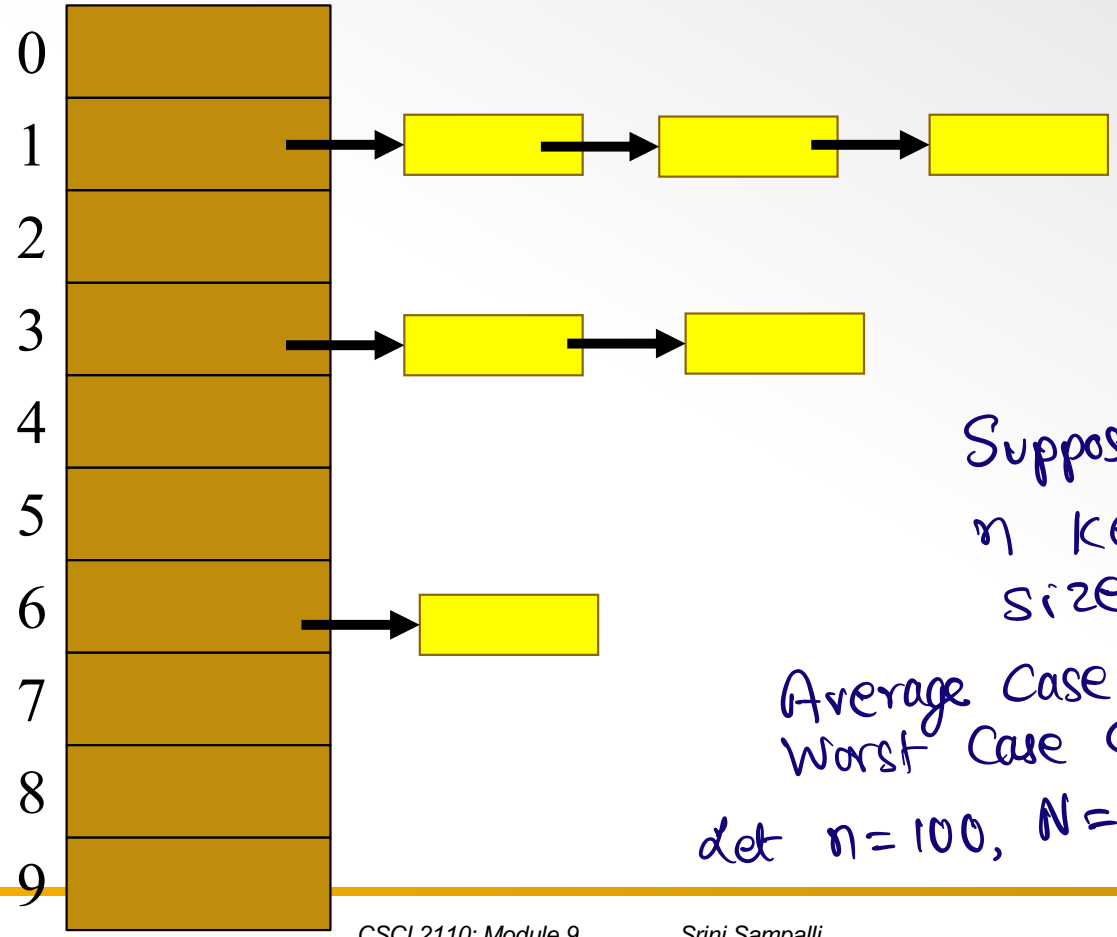
Assume that the size of the table is 5 and the hash function is defined as follows: add the positions of the word's letters in the alphabet (A=1, B=2, etc.) and compute the sum's remainder after division by 5.

SKILL	**= 19 + 11 + 9 + 12 + 12**	**= 63**
ABILITY	**= 1 + 2 + 9 + 12 + 9 + 20 + 25**	**= 78**
KNOWLEDGE	**= 11 + 14 + 15 + 23 + 12 + 5 + 4 + 7 + 5**	**= 96**
HARDWORK	**= 8 + 1 + 18 + 4 = 23 + 15 + 18 + 11**	**= 98**
ATTITUDE	**= 1 + 20 = 20 + 9 + 20 + 21 + 4 + 5**	**= 100**



63 mod 5 = 3
78 mod 5 = 3
96 mod 5 = 1
98 mod 5 = 3
100 mod 5 = 0

Separate chaining: An arraylist of linked lists - Search Complexity



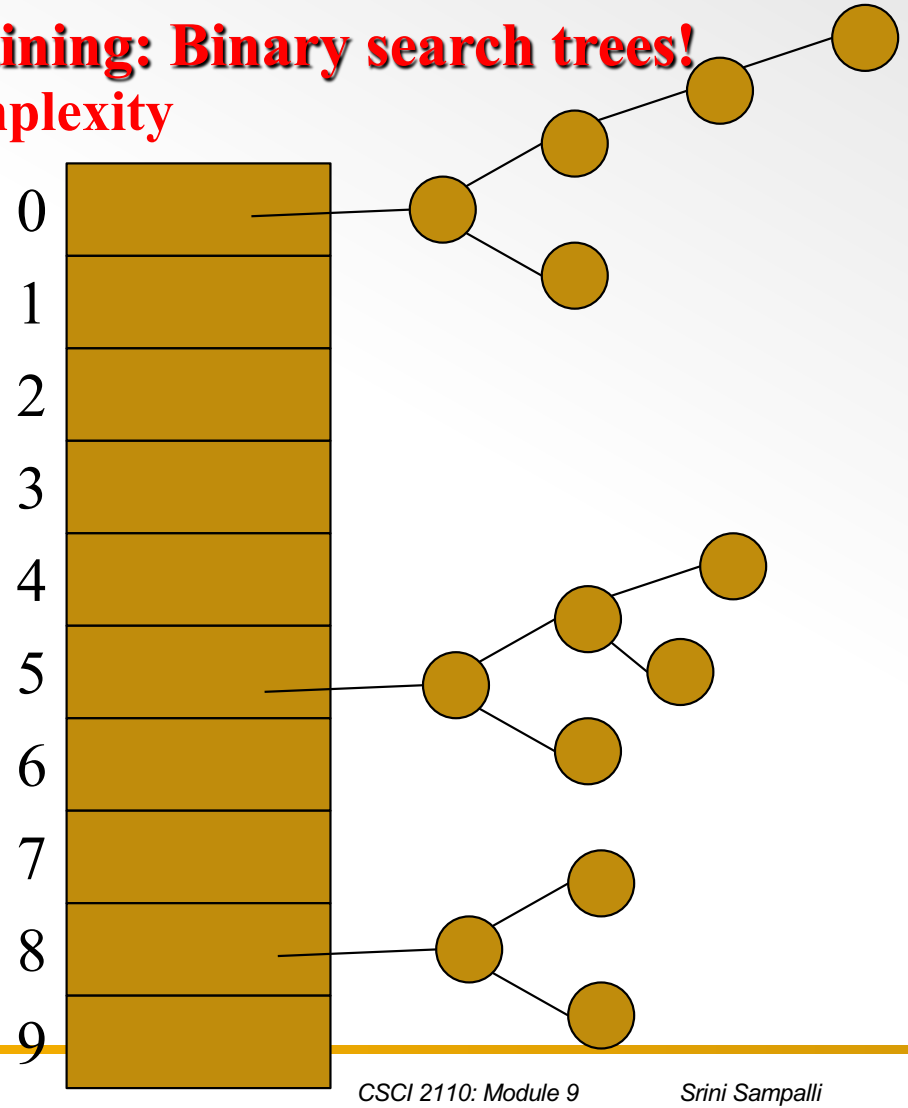
Suppose that there are n keys & the table size is N

Average Case Complexity $= O\left(\frac{n}{N}\right)$

Worst Case Complexity $= O(n)$

Let $n = 100$, $N = 10$, avg. # searches $= 100/10 = 10$

Separate chaining: Binary search trees!
- Search Complexity

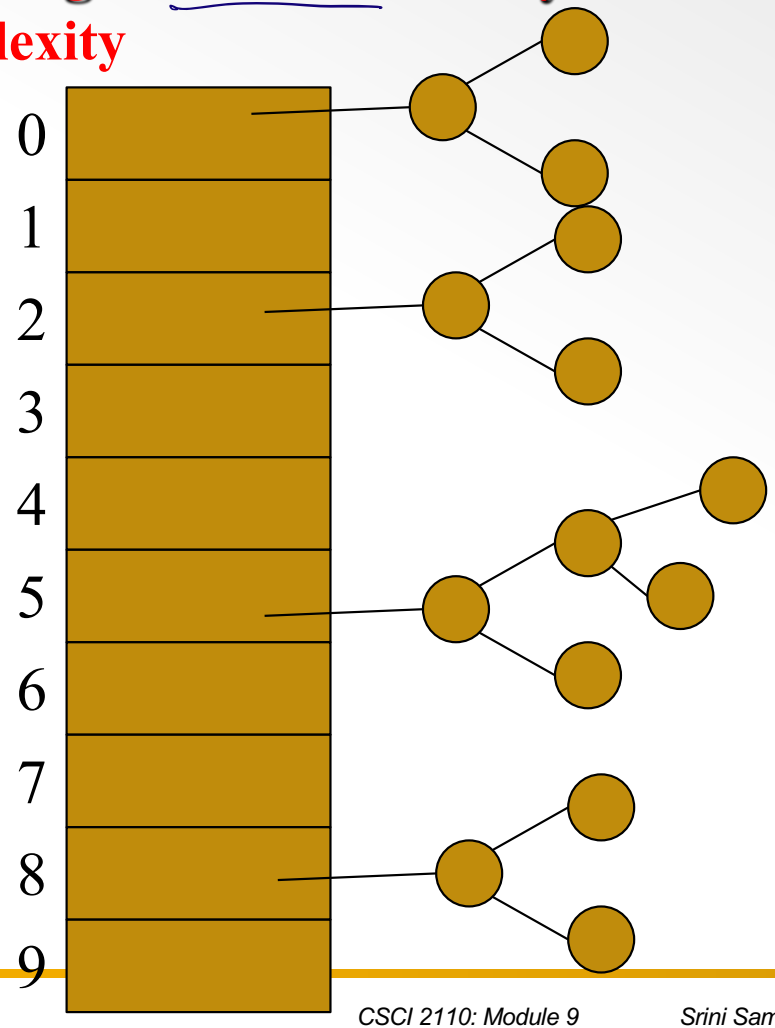


Avg. Case : $O\left(\frac{n}{N}\right)$

worst Case : $O(n)$

Separate chaining: Balanced Binary Search Trees!

- Search Complexity



Worst Case: $O(\log_2 n)$

Avg. Case: $O\left(\frac{\log_2 n}{N}\right)$

Inching closer to $O(1)$